Topology-Aware Mapping for Photonic Interconnects in TPU-Class LLM Accelerators

Technical Summary Sheet

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I. Key Insights and Innovations

1) **The local accelerator mapping is invariant under the photonic swap; the distributed schedule is what changes.** On the modeled 64-chip torus, the same partitioning, dataflow, and tile signature is selected across all seven interconnect technologies we evaluated. Out of 24 cases the inner mapping does not change once. The benefit lives one layer up: 5 of 16 workload and architecture pairs flip the optimal communication pattern, and the best alternative schedule (remote pooled-HBM streaming on KV attention with EVK100-class photonics) recovers about 73.9 % of the baseline schedule's energy. We report these as separate columns rather than collapsing them into a single "mapping changed" headline, because the distinction is exactly what the result is about.
2) **Multicast and distance-insensitive transmission, not raw bandwidth, drive the shared-state advantage.** A deliberately strong chiplet baseline at 3.5 Tb/s and 6 pJ/bit per hop is included specifically to defeat the bandwidth hypothesis. Even against that baseline, the M1000 photonic option reduces shared-DRAM multicast energy by about 99.85 %: N unicasts collapse to one logical broadcast, and contention disappears under a switched fabric. A distance-parameterized link model splits each technology into a conversion term and a per-millimeter transmission term so that electrical pJ/bit grows with system size and photonic pJ/bit stays roughly flat. A regression test enforces the crossover.
3) **Photonic interconnect is conditional, not universal.** On a 9216-chip Ironwood-class pod, the public Gemma 4 26B mixture-of-experts proxy reaches a $42.4\times$ inference latency speedup and a 94.05 % energy reduction with the full photonic compiler stack. The longer long-context Gemini-class proxy reaches $14.5\times$. Training, by contrast, reaches only $3.4\times$ and $1.4\times$ respectively, because once communication shrinks, the remaining time is dominated by matrix- multiply compute. Strong scaling exposes that compute and HBM ceiling; weak scaling shows where photonic multicast still pays. The strongest case for photonics in our setup is inference and long-context serving, not training payoff.

II. Contributions

Simone Surdi. Network and topology modeling: ring, mesh, and 3-D torus baselines along with the optical crossbar and switched-fabric abstractions; the logical cluster-memory extension to the accelerator architecture; the iterative network-aware feedback loop above the mapper; mapping-signature extraction; the cluster, TPU-cube, and strong chiplet-baseline studies; the technology catalog and the sensitivity sweeps over bandwidth and per-bit energy.

Pranav Sitaraman. Compiler and pod-scale analysis: the distance-parameterized link energy model; optical circuit switching and optical packet switching sensitivity points; the four named distributed-schedule choices (SPMD-style collective, remote pooled-HBM streaming, replicated cache, and photonic multicast cache fill); the 9216-chip Ironwood-class stage model spanning prefill, decode, KV/cache refill, mixture-of-experts dispatch, training, gradient synchronization, and optimizer state; strong and weak scaling on Gemma 4 anchors plus explicitly labeled long-context Gemini-class proxies; the ablation ladder across switching modes; the regression suite around the central physics; and the report, poster, technical summary, and presentation deliverables.

III. Reproduction and Validation

The full chain of evidence behind every quantitative claim in this sheet is reproducible end-to-end from a single project repository. Three command-line entry points cover regeneration of the per-stage data tables, end-to-end execution of the supporting analysis notebook, and rebuilding of the four submission documents (this technical summary, a four-page IEEE conference report, an A0 poster, and an accompanying slide deck). A regression suite enforces the central physical claims: the electrical-versus-photonic distance crossover, the inner-mapping invariance, the multicast scaling rule, the strong/weak scaling separation, and the optical-switching sensitivity band. Every figure in the submission package is regenerated from the checked-in evidence tables; there are no hand-edited plots.

IV. Scope

The model includes mapper action counts, per-chip memory cost, distance-aware link energy, hop count, contention, multicast semantics, optical switching, and a stage-level overlap term that prevents double-counting compute and network time. It does not include packet-level queue dynamics, thermal drift across the photonic stack, laser aging, area or cost analysis, model-quality effects, or the specific code-generation paths a production compiler would take. Those are explicitly listed as follow-on work rather than treated as hidden assumptions.